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Research and summary of SNR estimation algorithm based on cognitive radio

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Abstract

The cognitive ability of Cognitive Radio (CR) can adjust appropriate transmission parameters according to different channel conditions and adapt to dynamic communication environments. Channel quality evaluation is a key technology of adaptive communication systems, and its real-time evaluation results provide a direct basis for adaptive communication transmission scheme switching. So it is necessary to study the estimation algorithms of signal-to-noise ratio (SNR). Several SNR estimation algorithms based on additive white Gaussian noise channel (AWGN) are studied in this paper, analyze and compare through theoretical derivation and computer simulation. The simulation results of the computer MATLAB software show that the algorithm introduced in the article has a high estimation accuracy in the entire range of the SNR simulation interval of 0 to 15dB, and can accurately reflect the current channel conditions to facilitate the CR system adjusts the communication strategy in time, provides a certain reference basis for CR channel estimation decision-making.

Keywords: Cognitive radio (CR); channel quality evaluation; signal-to-noise ratio (SNR) estimation

1. Introduction

Dr. Joseph Mitola first proposed the concept of Cognitive Radio (CR) in the literature [1]. CR is an intelligent wireless communication system that can sense the electromagnetic environment it is in, and can dynamically and automatically adjust working parameters to adapt to the external environment [3]. Channel quality evaluation is one of the key technologies of adaptive communication, and the signal-to-noise ratio (SNR) is an effective evaluation

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index for wireless communication channel environment and communication quality. The estimated value of the SNR determines the communication parameters at the next moment, thereby improving the real-time and accuracy of the CR system's response to changes in the channel state.

2. Cognitive radio communication model

The channel feedback communication system based on CR is shown in Fig.1. Where a_n is the mapped signal and satisfies the condition $E\{a_{n1}a_{n2}\} = \sigma_a^2\delta(n_1 - n_2)$, h_k is the shaping filter coefficient, and N_{ss} is the number of upsampling points per symbol. When $\omega_c = 0$, it is the baseband signal transmission system model. When a_n is a BPSK signal, the channel is a real Gaussian white noise channel; when a_n is a high-order modulated complex signal such as QPSK, the channel is a complex Gaussian white noise channel.

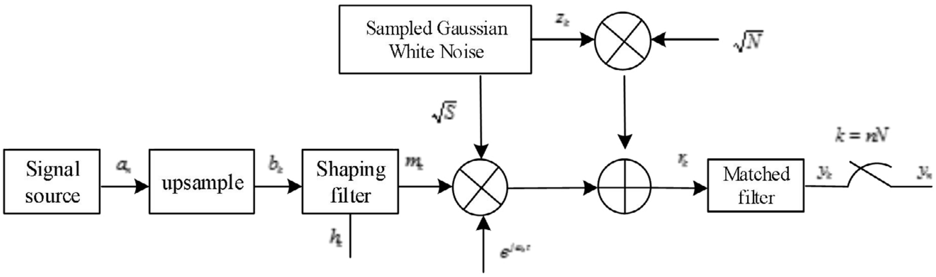

Fig. 1. Cognitive radio communication system model

The signal after oversampling and filtering can be expressed as

$$m_k = \sum_n a_n h_{k-nN_{ss}} \tag{1}$$

Where h_k represents the tap coefficient of the shaping filter.
The signal at the receiving end can be written as

$$r_k = \sqrt{S}m_k + \sqrt{N}z_k \tag{2}$$

Among them, Z_k is zero-mean normalized AWGN, the power of the signal is represented by S, and the power of the noise, is represented by N. The signal output from the receiving end can be written as

$$y_k = \left(\sqrt{S}m_k + \sqrt{N}z_k\right) \otimes h_{-k}^* = \sqrt{S}\sum_l h_l m_{k-l} + \sqrt{N}\sum_l h_l z_{k-l} \tag{3}$$

In the above formula, $\otimes$ denotes the discrete convolution operation, $*$ denotes the conjugate operation. Sampling the output signal to obtain

$$y_n = y_k |_{k=nN_{ss}} = \sqrt{S}a_n g_0 + \sqrt{N}w_n \tag{4}$$

Where

$$\begin{aligned} g_k &= h_k \otimes h_{-k}^* = \sum_l h_l h_{k-l} \\ w_n &= y_k \,|_{k=nN_{ss}} = \sum_l h_l z_{k-l} \,|_{k=nN_{ss}} \end{aligned} \tag{5}$$

3. Channel quality assessment method

Channel quality evaluation is a key technology of CR adaptive communication system, and its real-time evaluation results provide a direct basis for adaptive communication transmission scheme switching. The SNR value of a signal can reflect the change of noise in signal transmission and its influence on signal transmission. The SNR information is essential prior knowledge, and the accuracy of the SNR estimation algorithm will directly affect the selection of the transmission parameters. It can be seen that it is very necessary to study the estimation algorithm of SNR.

3.1. Maximum Likelihood (ML) algorithm for SNR estimation

The ML estimation algorithm [2] [6] is according to the maximum-likelihood estimation principle. Considering the complex signal, the complex channel received multiplexed signal can be expressed as

$$x(n) = x_I(n) + jx_Q(n) = \sqrt{P_s}(s_I(n) + js_Q(n)) + \sigma_v(v_I(n) + jv_Q(n)) \tag{6}$$

In the above formula, $s_I(n)$, $s_Q(n)$, $v_I(n)$ and $v_Q(n)$ represent the signal and noise of channel I and channel Q respectively. Assume that the noises of the I and Q signals have zero mean, and the variance is $\sigma_v^2/2$, respectively. And suppose that the noises of I and Q are independent of each other, P_s is the signal power scale factor.

Assuming that the signal and noise are independent of each other, the joint probability density function of N received signals can be obtained

$$\begin{aligned} f(X_I, X_Q \mid P_s, \sigma_v^2) &= \prod_{n=0}^{N-1} f(x_I, x_Q \mid P_s, \sigma_v^2) \\ &= \left(\frac{1}{\pi\sigma_v^2}\right)^N \exp[-\frac{1}{\sigma_v^2}[\sum_{n=0}^{N-1}(x_I(n) - \sqrt{P_s}s_I(n))^2 + \sum_{n=0}^{N-1}(x_Q(n) - \sqrt{P_s}s_Q(n))^2]] \end{aligned} \tag{7}$$

The likelihood function may therefore be expressed as follows

$$\begin{aligned} \Gamma\left(P_s, \sigma_v^2\right) &= \ln f(X_I, X_Q \mid P_s, \sigma_v^2) \\ &= -N\ln(\pi\sigma_v^2) - \frac{1}{\sigma_v^2}\left[\sum_{n=0}^{N-1}(x_I(n) - \sqrt{P_s}s_I(n))^2 + \sum_{n=0}^{N-1}(x_Q(n) - \sqrt{P_s}s_Q(n))^2\right] \end{aligned} \tag{8}$$

The maximum-likelihood estimate of P_s and σ_v^2 is determined by

$$\begin{aligned} &\frac{\partial}{\partial P_s}\Gamma(P_s, \sigma_v^2) = 0 \\ &\frac{\partial}{\partial \sigma_v^2}\Gamma(P_s, \sigma_v^2) = 0 \end{aligned} \tag{9}$$

Therefore

$$\hat{\rho}_{ML\ RxDA} = \frac{\hat{P}_{\text{sML}}}{\hat{\sigma}^2_{vML}} = \frac{N_{ss}{}^2 \left[\frac{1}{N} \sum_{n=0}^{N-1} \text{Re}\left\{ x^*(n)s(n) \right\} \right]^2}{\frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 - N_{ss} \left[\frac{1}{N} \sum_{n=0}^{N-1} \text{Re}\left\{ x^*(n)s(n) \right\} \right]^2} \tag{10}$$

Among them, Re{·} represents the real part of the complex signal, and the ML SNR estimation of the real signal only needs to replace the conjugate operation in the above formula with the corresponding real signal.

For real channels, multiply the denominator of equation (10) by $N/(N-3)$ to reduce the estimation deviation

$$\hat{\rho}_{ML\ RxDA,\text{real}} = \frac{\hat{P}_{\text{sML}}}{\hat{\sigma}^2_{vML}} = \frac{N_{ss}{}^2 \left[\frac{1}{N} \sum_{n=0}^{N-1} \text{Re}\left\{ x^*(n)s(n) \right\} \right]^2}{\frac{1}{N-3/2} \sum_{n=0}^{N-1} |x(n)|^2 - \frac{N_{ss}}{N(N-3/2)} \left[\frac{1}{N} \sum_{n=0}^{N-1} \text{Re}\left\{ x^*(n)s(n) \right\} \right]^2} \tag{11}$$

For complex channels, multiply the denominator of equation (10) by $N/(N-3/2)$ to reduce the estimation deviation

$$\hat{\rho}_{ML\ RxDA,complex} = \frac{\hat{P}_{\text{sML}}}{\hat{\sigma}^2_{vML}} = \frac{N_{ss}{}^2 \left[\frac{1}{N} \sum_{n=0}^{N-1} \text{Re}\left\{ x^*(n)s(n) \right\} \right]^2}{\frac{1}{N-3} \sum_{n=0}^{N-1} |x(n)|^2 - \frac{N_{ss}}{N(N-3)} \left[\frac{1}{N} \sum_{n=0}^{N-1} \text{Re}\left\{ x^*(n)s(n) \right\} \right]^2} \tag{12}$$

3.2. Second-and Fourth Order Moments (M2M4) algorithm for SNR estimation

The Second-and Fourth Order moments (M2M4) is a typical representative of statistics-based SNR estimation algorithms. In the AWGN channel, since the odd-order moment of the Gaussian random process is zero and the even-order moment is not zero, its high-order cumulant is always equal to zero [4][5].

According to the signal model defined in the previous section of this paper, since the mean value of signal and noise can be understood as zero, and the two are independent of each other, for the complex channel of high-order modulation mode, the second and fourth moments of the received signal are defined as follows

$$\begin{aligned} M_2 &= E[x(n)x(n)^*] = E[|s(n)|^2] + E[s(n)v^*(n)] + E[|v(n)|^2] + E[s^*(n)v(n)] = \mathrm{P}_s + \sigma_v^2 \\ M_4 &= E[(x(n)x^*(n))^2] = E[(s(n)+v(n))^2(s^*(n)+v^*(n))^2] = k_s P_s + 4P_s\sigma_v^2 + k_v\sigma_v^2 \end{aligned} \tag{13}$$

Where

$$k_s = \frac{E\left[|s(n)|^4 \right]}{E^2\left[|s(n)|^2 \right]} \tag{14}$$

And

$$k_v = \frac{E\left[|v(n)|^4 \right]}{E^2\left[|v(n)|^2 \right]} \tag{15}$$

Solving get an estimate of the signal and the noise power

$$\hat{S} = \frac{M_2(k_v - 2) \pm \sqrt{\left(4 - k_s k_v\right) M_2^2 + M_4(k_s + k_v - 4)}}{k_s + k_v - 4} \tag{16}$$

And

$$\hat{N} = M_2 - \hat{S} \tag{17}$$

For complex channels, $k_s = 1$ and $k_v = 2$; for real channels, $k_s = 1$ and $k_v = 3$.
Therefore

$$\hat{\rho}_{M2M4,real} = \frac{\hat{S}_{real}}{\hat{N}_{real}} = \frac{\frac{1}{2}\sqrt{6M_2^2 - 2M_4}}{M_2 - \frac{1}{2}\sqrt{6M_2^2 - 2M_4}} \tag{18}$$

$$\hat{\rho}_{M2M4,complex} = \frac{\hat{S}_{complex}}{\hat{N}_{complex}} = \frac{\sqrt{2M_2{}^2 - M_4}}{M_2 - \sqrt{2M_2{}^2 - M_4}} \tag{19}$$

In practical applications, the second and fourth moments of the complex signal and the real signal can be approximated by the following two formulas

$$\begin{aligned} \hat{M}_2 &\approx \frac{1}{N}\sum_{n=0}^{N-1}\left|x(n)\right|^2 \\ \hat{M}_4 &\approx \frac{1}{N}\sum_{n=0}^{N-1}\left|x(n)\right|^4 \end{aligned} \tag{20}$$

3.3. MMSE algorithm for SNR estimation

Assuming the sent training sequence is $\alpha = \{\alpha_1,\alpha_2,\ldots,\alpha_L\}$, the received vector signal is$\{\chi = x_1,\ x_2,\ldots,x_L\}$:, χ can be written as

$$x = f\alpha + v \tag{21}$$

Where $\nu = \{v_1,v_2,\ldots,v_L\}$ is the Gaussian noise vector signal; f is the attenuation factor, which is an unknown constant.

As for MMSE, the estimation error and the estimation are orthogonal to each other, the minimum mean square error estimate of f can be obtained

$$\begin{aligned} &\left(x - \hat{f}\alpha\right)\left(\hat{f}\alpha\right)^H = 0 \\ \Rightarrow \quad & x\alpha^H = \hat{f}\alpha\alpha^H \\ \Rightarrow \quad & \hat{f} = \frac{x\alpha^H}{|\alpha|^2} = \frac{C}{|\alpha|^2} \end{aligned} \tag{22}$$

In the above formula, the superscript H is the conjugate transpose of the matrix; $|\alpha|$ is the modulo operation, and $C = x\alpha^H$ is the correlation peak of the sequence. By estimating the attenuation factor $\hat{f}$, using the received signal and the known training sequence, the SNR expression is obtained

$$SNR = \frac{\left|\hat{f}\alpha\right|^2}{\left|x-\hat{f}\alpha\right|^2} = \frac{|C|^2}{|\alpha|^2 E - |C|^2} \tag{23}$$

Where $\mathrm{E} = |x^2|$ is the received signal energy.

4. Simulation results and analysis

Equation (24) gives the expressions of estimation error and normalized mean square error (NMSE). In this section, the performance of the three algorithms will be simulated and compared based on these two indicators.

$$\begin{aligned} Bias(\hat{\rho}) &= \frac{1}{M}\sum_{i=1}^{M}(\hat{\rho}_i - \rho) \\ NMSE(\hat{\rho}) &= \frac{1}{M}\sum_{i=1}^{M}\left(\frac{\hat{\rho}_i - \rho}{\rho}\right)^2 \end{aligned} \tag{24}$$

Where, M is the number of Monte Carlo simulations, $\hat{\rho}(i)$ is the estimated value of the i-th simulation, ρ is the true value of SNR.

The SNR estimator algorithms are investigated by the computer simulation of QPSK signals in AWGN, the roll-off coefficient of the root raised cosine filter is 0.3. The number of sampling points per symbol is $N_{SS} = 8$. The signal data observation length N is 1024, the corresponding SNR range is 0-15dB. For each algorithm, the estimated value of the SNR is calculated 50 times and averaged, the NMSE simulation results are shown in Fig. 2.

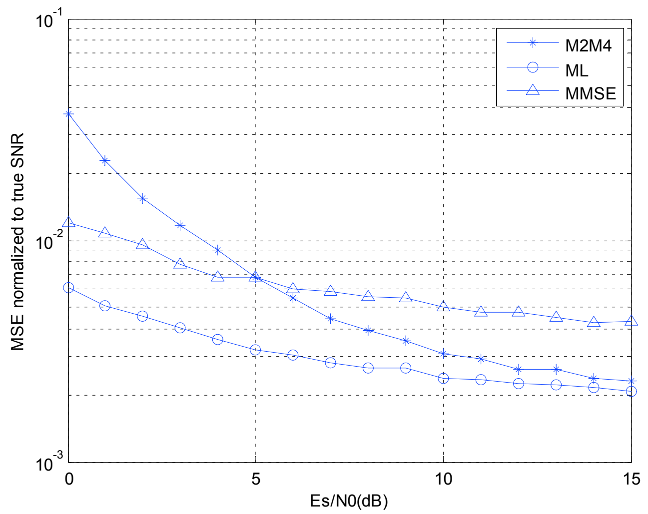

Fig. 2. Normalized mean square error of SNR

From the simulation results of Fig. 2, when the SNR is above 3dB, the NMSE of the estimated values of the three algorithms is less than 10^{-2}. However, the performance of the M2M4 algorithm decreases when the SNR is lower than 5dB. ML algorithm performance over the entire interval optimal SNR, indicating that the algorithm is robust to noise ratio is better. The MMSE method needs to send training sequences periodically. The magnitude of the estimation error of the SNR will directly affect the accuracy of the communication parameter selection, thereby affecting the performance of the CR communication system. The simulation results show that the above-mentioned algorithms have high estimation accuracy and can be used in CR systems.

5. Conclusions

CR system can adaptively adjust transmission parameters such as channel coding and modulation based on SNR feedback. Through theoretical deduction and computer simulation, the three algorithms introduced in the article have an estimated mean square error of less than 0.1dB in the interval of 0-15dB. In practical applications, the calculation complexity and algorithm performance should be considered comprehensively to select the appropriate algorithm.

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